

Junit



JAVA

Download and Install

Marc Philipp edited this page on Sep 29, 2023 · [35 revisions](#)

To download and install JUnit you currently have the following options.

Plain-old JAR

Download the following JARs and add them to your test classpath:

- [junit.jar](#)
- [hamcrest-core.jar](#)

Maven

Add a dependency to `junit:junit` in `test` scope. (Note: 4.13 is the latest stable version as of the latest edit on this page.)

```
<dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
```



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'Enclosed' test runner example

Aggregating tests in suites

Assertions

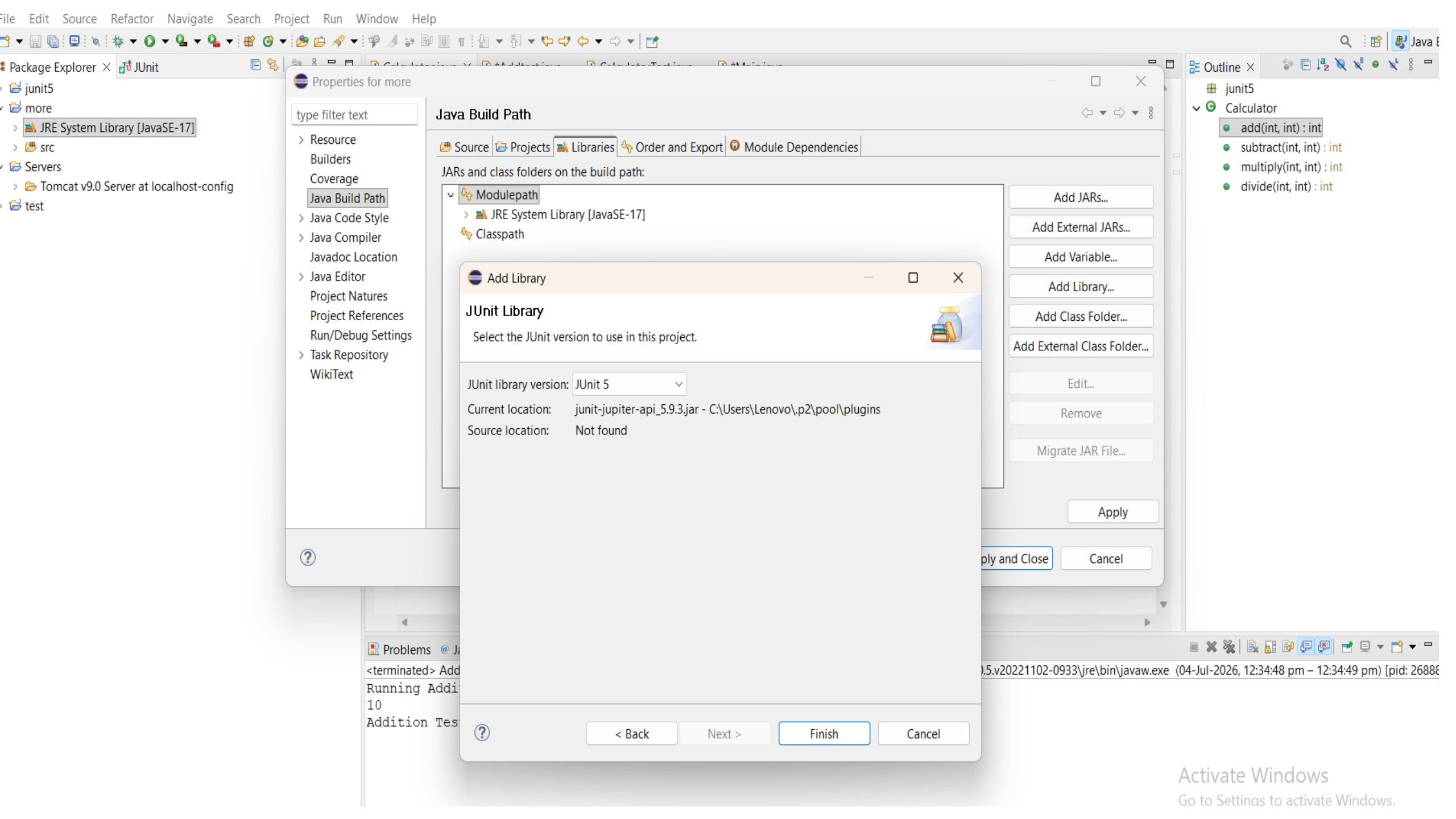
Assertions CN

Assumptions with assume

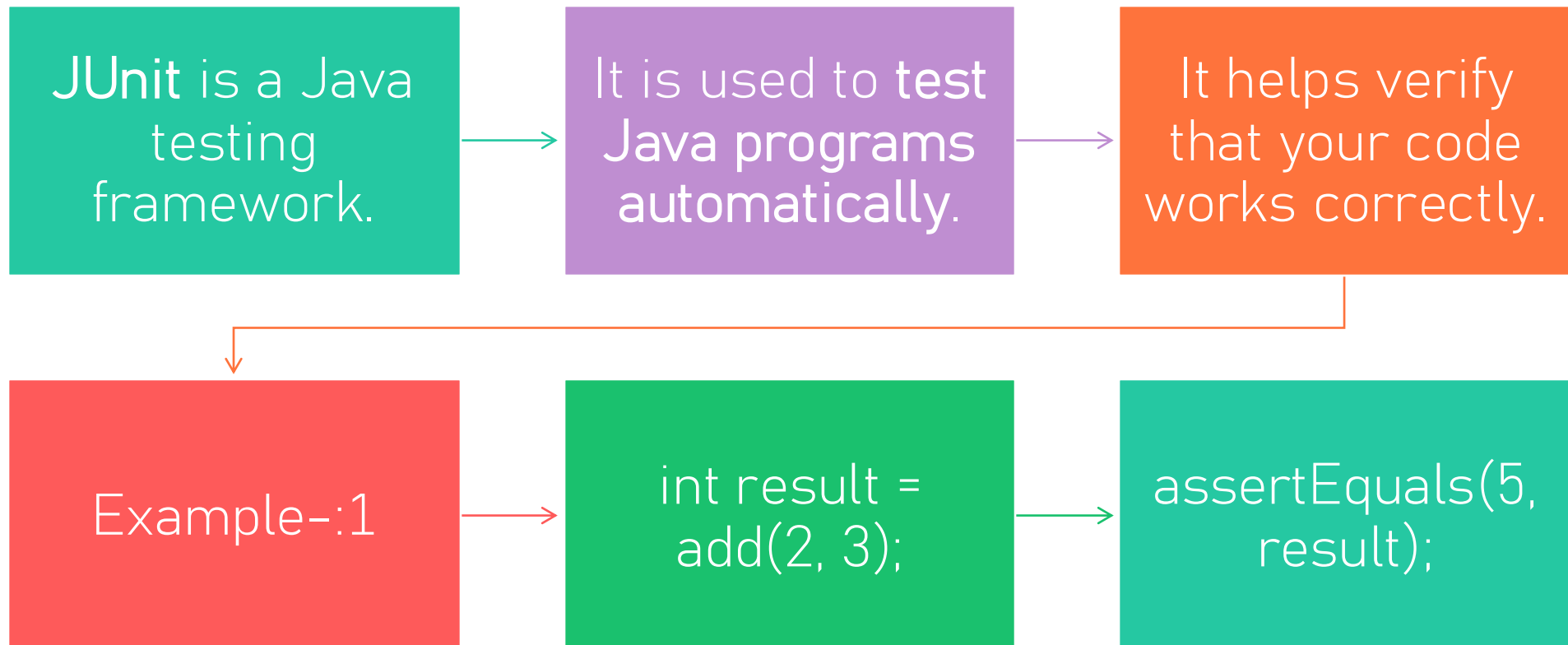
Categories

Continuous testing

Activate Windows
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What is JUnit?



JUnit Versions

JUnit 4	JUnit 5
Older version	Latest version
Package: org.junit	Package: org.junit.jupiter
Uses @Before and @After	Uses @BeforeEach and @AfterEach
Uses Assert class	Uses Assertions class
Limited features	More features like nested tests, dynamic tests, and better extension support

What is JUnit Jupiter?

JUnit Jupiter is the core programming and testing model of JUnit 5.

It provides:

Annotations like `@Test`, `@BeforeEach`, `@AfterEach`, `@BeforeAll`, and `@AfterAll`.

The Assertions class for checking test results.

Example:

```
import org.junit.jupiter.api.Test;
```

What is the purpose of Assertions?

The Assertions class is used to **verify** whether the expected result matches the actual result.

If the values match, the test passes; otherwise, it fails.

Common assertion methods:

`assertEquals()`

`assertTrue()`

`assertFalse()`

`assertNull()`

`assertNotNull()`

`assertThrows()`

Example:

```
assertEquals(100, totalAmount);
```

Why do we use `import static org.junit.jupiter.api.Assertions.*;`?

- This statement imports **all static assertion methods** from the **`Assertions`** class.
- It allows us to call assertion methods directly without writing the class name every time.
- Without static import:
 - `Assertions.assertEquals(10, result);`
`Assertions.assertTrue(result > 0);`
- With static import:
 - `assertEquals(10, result);`
`assertTrue(result > 0);`
- **Benefit:** It makes the code shorter, cleaner, and easier to read.

What is the difference between `@BeforeEach` and `@BeforeAll`?

`@BeforeEach`

Runs before every test method

Executes multiple times (once per test)

Usually used to create fresh test data

Method does not need to be `static`

`@BeforeAll`

Runs only once before all test methods

Executes only once

Usually used for one-time setup such as database or file initialization

Method must be `static` (unless using a per-class test instance lifecycle)

Example

```
@BeforeEach
```

```
void setup() {
```

```
System.out.println("Runs before every test");
```

```
}
```

```
@BeforeAll
```

```
static void initialize() {
```

```
System.out.println("Runs once before all tests");
```

```
}
```

What is the purpose of the `@Test` annotation?

The `@Test` annotation marks a method as a **test method**.

JUnit recognizes methods annotated with `@Test` and executes them automatically during testing.

Example:

```
@Test
void testLogin() {
    assertTrue(true);
}
```

Without the `@Test` annotation, JUnit will not recognize or execute the method as a test.

Quick Revision

Question

What is JUnit?

Why use JUnit?

What is JUnit Jupiter?

What is Assertions?

Why use `import static Assertions.*`?

What is `@Test`?

`@BeforeEach` vs `@BeforeAll`

JUnit 4 vs JUnit 5

Short Answer

A Java framework for unit testing.

To automate testing and improve code quality.

The testing API and programming model of JUnit 5.

A class used to verify expected and actual results.

To call assertion methods directly without `Assertions..`

Marks a method as a test method.

`@BeforeEach` runs before every test; `@BeforeAll` runs once before all tests.

JUnit 5 uses `org.junit.jupiter`, provides more features, and is the recommended version.

JUnit 4/JUnit 5

```
import org.junit.Test;
```

```
import static org.junit.Assert.*;
```

```
-----  
-----
```

```
import org.junit.jupiter.api.Test;
```

```
import static org.junit.jupiter.api.Assertions.*;
```

```
package more;

import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;

class Addtest1 {

    @Test
    void testAddition() {

        System.out.println("Running Addition Test");
        assertEquals(5, 2 + 3);
        System.out.println("Addition Test Passed");
    }
}
```